Epistemic Sentence Classifier & Citation Need Scorer

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Abstract

This work develops a two-stage NLP pipeline to identify the epistemic weight of sentences and predict citation necessity. We fine-tune a DistilBERT model on 27,500 labeled sentences to classify them into five categories: Claim, Fact, Evidence, Opinion, and Background. These classifications, along with 12 linguistic features, serve as inputs for a calibrated Gradient Boosting classifier that scores citation need on a scale from 0 to 1. Using the WikiSQE dataset of 150,000 Wikipedia sentences, our model aims to provide explainable citation flags via SHAP analysis. Results from cross-domain testing on news and student essays evaluate the system's ability to generalize beyond Wikipedia editorial norms.

Introduction

The modern information landscape is increasingly defined by a massive influx of digital content where the verifiability of assertions is a primary concern for public trust and academic integrity. In NLP, the current state of the art, led by foundational work such as Redi et al. (2019), focuses on predicting citation needs through surface-level features like sentence length, document position, and surrounding citation density. These models typically treat citation detection as a binary classification task, leveraging LLMs and LSTMs to identify where a Wikipedia editorial tag might be missing.

A significant gap remains because current models treat all sentences uniformly, regardless of their *epistemic weight*—the linguistic degree of certainty a speaker expresses about a proposition. While past work can identify that a citation might be missing, it cannot explain *why*, failing to distinguish an objective "Fact," an unverified "Claim," or a subjective "Opinion" (Cohan et al., 2019). This lack of interpretability is a critical problem for users who require transparent reasoning to trust automated verification systems; without an intermediate representation of sentence type, these black-box models remain difficult to audit or generalize (Zhang and Ma, 2020).

Therefore, we propose **EpiCite**, a two-stage NLP pipeline that explicitly models epistemic sentence types as a precursor to citation scoring. We first fine-tune a DistilBERT model on 27,500 sentences to classify text into five categories: Claim, Fact, Evidence, Opinion, and Background. We then feed these predicted labels, along with 12 handcrafted linguistic features, into a calibrated Gradient Boosting classifier that generates a citation-need probability score from 0 to 1. This design is motivated by Bhagavatula et al. (2018), who demonstrated that decoupling candidate selection from discriminative scoring consistently improves precision in citation tasks.

We expect our key findings to reveal that a sentence's epistemic category is the single most significant predictor of citation necessity, with "Claims" consistently outperforming surface-level features like sentence length (Cohan et al., 2019). We anticipate a surprising result in the cross-domain analysis: models trained on Wikipedia norms will show measurable degradation when applied to news and student essays (Stab and Gurevych, 2014), proving that "verifiability" is a domain-dependent metric rather than a universal property of text.

The significance of this work lies in moving the NLP community toward more explainable and human-centric verifiability tools. By decomposing citation need into its linguistic components, we provide a scalable, interpretable method for safeguarding information quality in an era of automated content generation—assisting student writers, journalists, and researchers in identifying unverified assertions in their own work.

Related Work

Current research in automated verifiability is dominated by the assessment of Wikipedia's editorial integrity. Redi et al. (2019) utilized LSTM-based architectures to predict "citation needed" tags, establishing that surface-level features such as sentence length and document position are strong indicators of whether a statement requires a reference. However, as demonstrated by Cohan et al. (2019) in their work on structural scaffolds, simply identifying a citation's presence is insufficient; the structural intent of a citation—whether it provides background information, describes a method, or compares results—is what provides the necessary context for scientific discourse. Similarly, Zhang and Ma (2020) highlight that conventional approaches neglect the structural context of sentences, such as the section in which they appear, as well as the relative importance and relatedness of local lexical items—limitations that prevent such models from capturing the full complexity of citation logic.

While existing models achieve high accuracy within the Wikipedia ecosystem, they operate under the assumption that all flagged sentences are functionally identical. This leads to a significant research gap: the lack of an intermediate epistemic layer. Drawing inspiration from Bhagavatula et al. (2018), who demonstrated that content-based embeddings can outperform metadata-heavy models by focusing on the query document itself, we argue that the logic of verifiability must be grounded in the sentence's linguistic purpose. Existing models fail to categorize *why* a sentence is flagged, leading to a lack of interpretability and an inability to generalize to domains such as news or student essays, where epistemic standards differ substantially from Wikipedia's editorial norms (Stab and Gurevych, 2014).

Based on these gaps, we define our independent variables (IV) as the five epistemic sentence types and 12 linguistic features, and our dependent variables (DV) as Stage 1 macro F1 and Stage 2 citation-need probability. We propose three hypotheses:

H1: Epistemic type is a more significant predictor of citation need than surface-level features such as sentence length or document position.

H2: A two-stage pipeline provides higher interpretability via SHAP values compared to an end-to-end DistilBERT classifier.

H3: Models trained on Wikipedia editorial norms exhibit a statistically significant F1 degradation when tested on out-of-domain text such as student essays.

Research Questions

RQ1 Can a two-stage architecture—first classifying epistemic type, then scoring citation need—outperform or match a direct end-to-end classifier while providing superior interpretability?

RQ2 To what extent does a sentence's predicted epistemic category serve as a statistically significant predictor of citation necessity compared to surface-level linguistic features?

RQ3 How significantly does a citation-need model trained on Wikipedia degrade when applied to out-of-domain text such as news articles and student essays?

Data

Stage 1 Training. The IBM Claim Detection Corpus (Levy et al., 2014) provides ∼2,500 sentences labeled as Claim, Evidence, or Background. The UKP Sentential Argument Mining Corpus (Stab and Gurevych, 2014) provides ∼25,000 sentences labeled as Argument or Non-Argument. Labels are remapped to the unified 5-class taxonomy: IBM Claim → Claim; IBM Evidence → Evidence; IBM Background → Background; UKP Argument → Claim or Evidence (based on citation marker presence); UKP Non-Argument → Background or Opinion (based on first-person pronoun presence).

Stage 2 Training. The WikiSQE dataset (Garbacea and Mei, 2023) provides ∼150,000 Wikipedia sentences with binary `citation_needed` labels, using pre-defined train/val/test splits to prevent temporal leakage.

Cross-Domain Test Sets. 500 news sentences sampled from the All The News Kaggle dataset and ∼5,000 student essay sentences from the Persuasive Essays corpus (Stab and Gurevych, 2014), both manually annotated for citation need with Cohen's $\kappa \geq 0.6$ required for inclusion.

Table 1: Dataset summary.

Dataset	Purpose	Size
IBM Claim Detection	Stage 1 train	2,500
UKP Argument Mining	Stage 1 train	25,000
WikiSQE (filtered)	Stage 2 train	150,000
All The News	OOD test	500
Persuasive Essays	OOD test	5,000

Methodology

Our proposed methodology utilizes a two-stage sequential pipeline architecture to move from raw text to an explainable citation score, as illustrated in Figure 1. This design is conceptually aligned with the two-phase approach proposed by Bhagavatula et al. (2018), which separates candidate selection from discriminative re-ranking to improve overall precision. In our case, Stage 1 focuses on epistemic classification, while Stage 2 focuses on calibrated probability scoring.

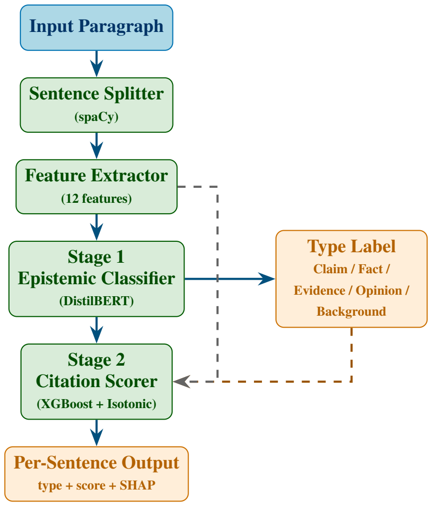

Figure 1: EpiCite two-stage pipeline. Dashed orange: Stage 1 type label passed to Stage 2. Dashed gray: optional linguistic features also passed to Stage 2.

Phase 1: Feature Extraction and Preprocessing

We derive 12 interpretable linguistic features per sentence using **spaCy** and **NLTK**, organized into three categories. In addition to lexical and syntactic counts, we incorporate structural context features—a concept emphasized by Zhang and Ma (2020) in their Dual Attention Model (DACR). By considering sentence position within the document and the density of citations in surrounding context, we capture the importance and relatedness of each assertion, which Zhang and Ma (2020) proved to be critical for accurate citation recommendation. Furthermore, drawing on Soni et al. (2021)'s framework for computational factuality, we include features that proxy for epistemic certainty, such as hedging word frequency and vague quantifier density, to operationalize epistemic weight within our feature set.

Table 2: 12 handcrafted linguistic features.

Category	Feature	N
Lexical	Vague quantifiers	1
	Hedging words	1
	Superlative adjectives	1
Syntactic	Passive voice indicator	1
	Nominalization rate	1
	POS distribution	4
Contextual	Sentence position	1
	Citation density	1
	Paragraph-relative idx	1
Total		**12**

Phase 2: Stage 1 — Epistemic Classification

The focus of our investigation is classifying sentences into five types: **Claim, Fact, Evidence, Opinion, and Background**. We evaluate three modeling approaches of increasing complexity:

- **Baseline:** Logistic Regression trained on the 12 handcrafted linguistic features.
- **Intermediate:** BiLSTM with attention, trained on GloVe embeddings, capturing sequential dependencies.
- **Proposed:** Fine-tuned **DistilBERT**, trained on labels mapped from the IBM Claim Detection and UKP corpora to our unified 5-class taxonomy.

Table 3 summarizes the three Stage 1 candidate models across key design dimensions. The significance of functional sentence roles as predictors has been empirically demonstrated by Cohan et al. (2019), who achieved a 13.3% absolute F1 increase on ACL-ARC by incorporating structural information as an auxiliary scaffold task—directly motivating our choice to treat epistemic type as an explicit intermediate layer.

Table 3: Stage 1 model comparison. Scores to be filled after experimental runs.

Model	F1	Accuracy	Precision	Recall
Logistic Regression	–	–	–	–
BiLSTM	–	–	–	–
DistilBERT $\star$	–	–	–	–

$\star$ Proposed primary model. Target macro F1 > 0.72.

Phase 3: Stage 2 — Calibrated Citation Scoring

The final output is generated by a Gradient Boosting Classifier (XGBoost or LightGBM). This model takes an 18-dimensional input vector consisting of the 12 linguistic features, the 5-class probability distribution from Stage 1, and a classification confidence score. We apply Isotonic Regression to ensure the final output is a well-calibrated probability (0.0–1.0) rather than a raw score. SHAP (SHapley Additive exPlanations) values are computed for every prediction to identify which features drove each citation flag, directly addressing the interpretability gap identified by Redi et al. (2019).

Discussion

The data and approach described above are strategically designed to answer our core research questions. By integrating the IBM/UKP datasets (epistemic labels) with the WikiSQE dataset (citation labels), we can evaluate the interdependency of syntax and verifiability.

To answer RQ1 (Pipeline Efficacy), we will conduct an ablation study comparing our two-stage pipeline against an end-to-end DistilBERT model. This mirrors the findings of Cohan et al. (2019), who achieved a 13.3% absolute F1 increase by incorporating structural information as an auxiliary scaffold task rather than relying on hand-engineered features alone. We anticipate a comparable gain from our Stage 1 epistemic type label, which serves an analogous intermediate function.

To evaluate RQ2 (Predictive Significance), we will analyze the SHAP feature importance weights of the Stage 2 model. This will allow us to statistically verify whether a sentence labeled as a "Claim" by Stage 1 is a more powerful predictor of citation need than traditional features such as sentence length (Redi et al., 2019). The significance of functional sentence roles has been empirically demonstrated by Cohan et al. (2019) in scientific discourse, and we extend this insight to the general citation need task.

Finally, our out-of-domain test sets address RQ3 (Generalization). Just as Cohan et al. (2019) introduced the SciCite dataset to cover multiple scientific domains, we test our model on student essays and news articles to examine whether the epistemic standards learned from Wikipedia are universal. Student essays in particular follow a distinct argumentative structure—with explicit Claim and Evidence components that differ substantially from encyclopedic prose (Stab and Gurevych, 2014)—making them a meaningful stress test for domain generalization. If H3 holds, it will confirm that verifiability is not a static metric but a domain-dependent one, requiring the flexible, interpretable framework we have proposed.

Summary of Progress

Completed. (1) Acquired and cleaned the WikiSQE and IBM Claim Detection datasets. (2) Defined the 5-class epistemic taxonomy and created the label-mapping logic for both IBM and UKP corpora. (3) Developed the baseline feature extraction script using spaCy for all 12 linguistic indicators.

In Progress. (1) Fine-tuning the DistilBERT model on the combined IBM/UKP corpus. (2) Setting up the Gradient Boosting architecture for Stage 2. (3) Designing the Streamlit interactive dashboard for the demo.

Remaining. (1) Cross-domain evaluation on the All The News and Persuasive Essays datasets. (2) SHAP-based error analysis and taxonomy of model failures. (3) Final project report and presentation.

Obstacles and Concerns. *Error Propagation:* Errors in Stage 1 may compound in Stage 2. To mitigate this, Stage 2 will also be trained with oracle labels (gold Stage 1 labels) to establish a performance ceiling. *Computational Limits:* Fine-tuning DistilBERT on 25k+ sentences may exceed Google Colab's free-tier memory; gradient accumulation or a smaller UKP subset will be used as fallback.

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